Predicting Glanders Disease Across a Lifestyle or Geographical Area

Chained Multi-Level Machine Learning on Four Unused Datasets

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What this project does, in short. Glanders is a dangerous bacterial disease of horses and donkeys. We asked one question: can we predict, using only public data, which countries will report glanders next year?

We combined four data sources that had never been used for this disease: indexed research papers, farm animal counts, World Bank statistics, and NASA climate records. We then trained six machine learning models and tested them on future years and on countries the models had never seen.

The result is a working early-warning model. It ranks likely countries about ten times better than random guessing, and its stated probabilities turned out to be trustworthy. Warning only about 6% of the world each year would catch roughly half of all reported outbreaks. The honest limitations are that the model works less well in countries it has not seen, and that it largely learns *which countries report* disease rather than *where* disease is. Both points are discussed openly in Section 9.

1 What We Tried To Do

Glanders is a bacterial disease that kills horses, donkeys and mules, and it can pass to people who handle infected animals. Most Indian equids belong to poor families who use them for transport, farming and construction. A district officer who knew a year ahead that glanders was likely could warn owners in time to act. So the question is:

> *Given what we know about a country this year, can we predict whether glanders will be reported there next year?*

One row in our data is one country in one year. The model returns a number between 0 and 1, which we read as a risk score. A score of 0.30 means roughly a 30% chance that glanders will be reported in that country next year.

The model does not diagnose an animal and cannot tell a farmer whether their horse is sick. It works at the level of countries and years, so it suits planning work. It is not a substitute for a veterinarian looking at a sick animal.

2 Related Work

Glanders. A neglected zoonosis of equids with high mortality and real human risk [1, 2]; a 2026 meta-analysis consolidates seroprevalence and names management and movement as the main drivers [3]. Brazil keeps an active control literature [4].

India. Outbreaks 2006–11 [7], serosurveillance 2015–18 [6], and a multi-state survey 2019–23 [5] establish *where* glanders is. None builds a predictive model.

Geographic risk. Spatiotemporal cluster detection for Brazilian glanders [8] and niche models for livestock disease [9] are both descriptive: clusters are found after the fact, not forecast.

ML for animal disease. The closest template is avian influenza: ARIMA versus random forest [11], ML-targeted surveillance [12], dynamic farm-level risk [13], and an outbreak model for Kuwait [10]. That template exists for influenza, not glanders.

The gap. The only ML on *B. mallei* is spectroscopic identification of the organism [14, 15], which is a laboratory test rather than a population forecast. Text mining has precedent in epidemic intelligence [16, 17], but not for glanders.

Methods. Random forests [20], XGBoost [18] and LightGBM [19], explained with SHAP [21]. Rare events call for precision-recall [22, 23]; probabilities are calibrated [25, 26]; validation respects temporal and spatial dependence [24]. Data: WAHIS [33], India's 2025 action plan [34], FAOSTAT [35], World Bank [37], NASA POWER [36], Europe PMC [38].

Position. Prior glanders work is descriptive or laboratory-based, prior veterinary ML is built for influenza, and text-mined occurrence has never been used for glanders. This study joins all four and reports what predicts and what does not.

3 Dataset Used

We used four public data sources. None had been used for glanders before, which is the main reason this counts as new work.

Table 1: The four data sources. Glanders events in Europe PMC were extracted by matching a country name and an event term within the same sentence, which yielded 263 events across 2000 to 2024.

Source	What it gives us	Size
Europe PMC (publications)	Where and when glanders was reported	6,674 papers → 263 events
FAOSTAT	How many horses, donkeys, mules, cattle, goats and sheep each country has	68,449 records, 1961–2023
World Bank	Money, farming, population, internet access	10 indicators, 217 countries
NASA POWER	Temperature, rainfall, humidity, wind	6 measures, 211 locations

Why these four sources. They cover what decides whether a disease can appear at all: whether it has been there before, how many animals could catch it, and whether the climate suits it. World Bank data adds a rough measure of how well a country detects and reports disease. Pulling several kinds of data together is what makes this stronger than an analysis of one source.

The assembled panel. All four sources are keyed on ISO-3 country code and calendar year, so they join without ambiguity. The full panel has 6,700 country-year rows covering 268 countries from 2000 to 2024. For modelling we keep only equine-holding countries, meaning those with a maximum stock of at least 1,000 equids, the years 2005 to 2023, and rows where next year's answer is observable. That leaves **3,059 rows, 44 candidate columns and 172 positive labels**.

Preprocessing. Five steps, in this order.

1. **Name reconciliation.** Country and area names were matched across sources through `pycountry` with explicit overrides, because the same country is written "Türkiye" in one file and "Turkey" in another. Without this the joins would silently drop rows.
2. **Aggregate removal.** World Bank regional and global totals were dropped by requiring a three-character ISO code, so rows such as "World" and "South Asia" cannot enter the panel as if they were countries.

3. **De-duplication.** Literature events were de-duplicated on the key `(iso3, year, record_id)`, so one paper cannot inflate a cell by being counted several times.
4. **Missing values.** Filled with the column median, and every column keeps a missing-indicator flag so the model can tell an imputed value from a real one. The fill is fitted inside the training pipeline, never on the full table. Missing data was rare: 0.99% overall and 5.9% in the worst column.
5. **Absence handling.** A country-year is coded 0 only after that country passes the equine-stock filter, so a blank is never read as proof that disease was absent. This is the step that keeps the label honest.

Feature engineering. Four groups of columns were built from the raw sources, all using information available in the current year or earlier. History columns are lags, a five-year running total and years since the last event. Population columns add derived quantities: equid share of all farm animals, equine density, and growth over one and three years. Climate columns pair each measure with an anomaly, its deviation from that country's own expanding mean. Section 5 lists all four groups.

Feature selection. Every input was screened for association with the answer and for duplication of another input, but the screens were read rather than obeyed. Only one column was dropped outright: total equids, because it is exactly the sum of horses, asses and mules and a tree can rebuild it. Four columns scored below both association thresholds and three near-duplicate pairs were flagged, yet all were kept, for the reason covered in Section 4: the univariate screen disagrees with what the fitted model actually uses, and one flagged column turns out to be among the three most important. No principal component analysis was used either; the limit here is signal rather than dimensionality.

4 Dataset Analysis

We start with the label, then move to how each input relates to it and how much the inputs duplicate each other.

The answer key. For every country and year the answer is yes or no: was glanders reported the following year? Only 172 of the 3,059 rows are yes, which is 5.6%.

Table 2: Numbers describing the answer key.

Measure	**Value**
Total rows	3,059
"Yes" rows (event next year)	172
"No" rows	2,906
Percentage "yes"	5.6%
Ratio of no to yes	about 17 to 1
Countries with at least one "yes"	56 of 161
Countries with "yes" in two or more years	30
Share of all "yes" in the top 5 countries	35.5%

What the distribution tells us. Three things follow from that table, and each one changed a decision later. First, yes is rare enough that always answering no would be right 94.4% of the time, so accuracy is useless here and we need scores built for rare events. Second, the disease comes back: 30 of the 56 positive countries were positive in two or more years, so past activity is a fair input rather than a shortcut. Third, the yes answers cluster, with five countries holding 35.5% of them, and those five are mostly rich countries that run publishing laboratories.

For that reason the target is termed *documented activity* rather than *disease occurrence*. A negative label can reflect either the absence of disease or the absence of a report.

Before training anything we measured how each input relates to the answer. Three measures were used on the numeric columns. *Spearman's* ρ checks whether two quantities rise and fall together, on a scale from -1 to

+1. *Mutual information* catches relationships that are not a straight line, and equals zero when a column tells you nothing about the answer. *Cramér's V* does the same job for text categories such as region.

Statistical association. The strongest monotone signal in the whole dataset is `years_since_last` at $\rho = -0.35$. The sign is negative because more years since the last outbreak means lower risk. This is the one broadly epidemiological result in the panel, and it says what we would hope: recent activity predicts future activity. On the same scale, `internet_pct` scores only +0.09 and `temp_mean_anom` about zero. For the two text categories, region gives Cramér's V of 0.243, which counts as a small association, and income band gives 0.044, which is negligible. Neither is strong enough to act as a stand-in for the label.

Correlation and redundancy. Several columns measure nearly the same thing. Variance inflation factor is the standard measure of that, where a value above 10 is usually treated as a problem. It reaches 227 for `temp_mean`, because annual mean, minimum and maximum temperature are near-duplicates of each other at country resolution. Three groups sit at $|\rho| \geq 0.90$: `occ_lag1` with `occ_count_lag1`, `cattle` with `log_livestock`, and `temp_mean` with `temp_min`. We kept one member of each group. This matters for reading coefficients in a linear model, and not much for trees, since a tree splits on thresholds and does not care whether two columns carry overlapping information.

Where the two importance measures disagree. This is the result that shaped the whole feature set. We compared how much each column correlates with the answer on its own with how much the trained model relies on it, using SHAP. The two rankings agreed almost not at all, with a rank correlation of -0.04 and only 5 of the 10 top features shared. Longitude is the clearest case: it ranks first on simple correlation and around twentieth in what the model uses. The explanation is that longitude does not cause glanders. It identifies which part of the world a country is in, and some parts publish more papers. Ranking features by marginal correlation would therefore favour columns that track reporting practice rather than disease occurrence. Features were selected on the conditional evidence instead, and correlation was used only to discard columns with no signal.

Table 3: Top 8 features by importance to the trained model, with marginal statistics alongside. The gap between the two sets of numbers is the disagreement described above.

Rank	Feature	Importance (mean |SHAP|)	Correlation (Spearman ρ)	Mutual info (MI)	Redundancy (VIF)
1	pop_total	1.457	+0.28	0.0639	10
2	internet_pct	0.892	+0.09	0.0000	5
3	log_livestock	0.702	+0.26	0.0571	3
4	goats	0.688	+0.19	0.0421	11
5	temp_mean_anom	0.635	-0.01	0.0006	2
6	precip_cv	0.579	-0.01	0.0008	2
7	humidity_anom	0.494	+0.03	0.0065	2
8	pop_density	0.473	+0.05	0.0262	2

What we removed, and why. Four columns had mutual information below 0.002 and $|\rho|$ below 0.02, which means no measurable relationship to the answer at all. We dropped one column that was exactly the sum of three others, since it adds nothing a tree cannot build itself. We dropped one member of each near-duplicate pair listed above. No principal component analysis was used: with the retained columns and 3,059 rows the limit is signal rather than dimensionality, and the section on results confirms that fewer columns generalise better across countries, so compressing further would have cost interpretability for nothing. Scaling is applied only to the logistic baseline, because tree models are indifferent to it.

5 What We Gave The Model (Features)

A feature is an input column. Forty-four were engineered and fourteen are used, all of them built from information available in the current year or earlier. The candidates fall into four groups.

- **Disease history (7 columns).** Outbreaks in the previous two years, the five-year running total, years since the last event, and world and regional activity shares for the same year. The disease persists once introduced, so this group carries the epidemiological prior.
- **Animal population (14 columns).** Log equid and total livestock stocks, the equid share of all farm animals, equid density, growth over one and three years, and the seven species counts. More animals means more chances to find the disease.
- **Climate (10 columns).** Mean, maximum and minimum temperature, precipitation total and its coefficient of variation, relative humidity, and wind, each paired with an *anomaly*. An anomaly is the difference from that country's own expanding mean, which separates "unusually wet for here" from an absolute number that mostly encodes latitude.
- **Country context (11 columns).** Income band, agricultural land share, agricultural value added, livestock production index, GDP per person, rural share, internet penetration, fertiliser intensity, life expectancy, population and density. These mostly measure how well a country detects disease.

Three derived columns are worth naming. *Equid share* is horses plus donkeys plus mules divided by all farm animals, which shows how much of the farming depends on equids. *Livestock units* counts cattle and camels at full weight and small ruminants at one tenth, following standard metabolic weights, so one number expresses grazing pressure across species. *Years since last event* uses 99 to mean "never recorded".

6 Models We Tried

We trained six models, including a deliberately simple one to compare against. The table gives each model's family, what it does, and the settings the tuning search picked.

Forest models train many trees *in parallel* on random subsets and average them, which reduces variance. Boosted models train trees *in sequence*, each fitting the error left by the last, which reduces bias. That difference explains why the two families end up with similar scores.

We left neural networks out because the data cannot support them. With 172 positives and 44 inputs, a network has far more settings to learn than it has examples to learn from, so it would memorise the training rows and fail on new ones. That is an argument from the data rather than a judgement about the method.

The collinear columns noted in Section 4 needed no special handling, since trees split on thresholds.

How the final model was chosen. On performance in countries it had never seen, because that is how the model would be used. Extra Trees came out best at 0.363. Random Forest scored higher on the easier time-based test, but the country test is the one that speaks to deployment.

We did not stack the models into a meta-learner. The gap between families is smaller than the uncertainty on any single score, so a meta-learner trained on 172 positives would add variance without a gain we could defend. We spent that effort on calibration instead, since that is what a threshold-based alert depends on.

7 How We Trained And Tested It

Several measures were taken so that nothing from the test period could influence training.

We ran three tests, and each answers a different question. They disagree for an informative reason: the country split always scores lower than the time split, and that gap is the price of meeting a country the model has never seen.

A random split would put the same country in both training and testing, which lets the model memorise country patterns and look good for the wrong reason. Grouping by country prevents that. The scores are defined

Table 4: The six models, their family, and the settings chosen by tuning.

Model	Family	What it does, and the settings chosen
Logistic Regression	linear	Adds the inputs with fixed weights and squashes the total into a 0–1 score. The honest baseline: if the complex models cannot beat it, the complexity is not worth it. Tuned settings: $C = 0.01$, ℓ_2 *penalty, balanced class weights*
Decision Tree	tree	A flowchart of yes/no questions, for example "horses > 10,000? then last outbreak < 2 years ago?". Easy to read, but one tree is weak, which is part of why we included it. Tuned settings: *depth 4, minimum leaf size 20, split criterion gini*
Random Forest	bagged trees	500 trees, each trained on a random part of the data, then they vote. Fixes the single tree's habit of memorising. Tuned settings: *depth 6, minimum leaf size 10, features per split 0.3*
Extra Trees	bagged trees	Same idea, with splits chosen more randomly. Extra randomness helps when examples are scarce, as they are here. Tuned settings: *depth 6, minimum leaf size 3, features per split square root of the column count*
XGBoost	boosted trees	Trees built one after another, each fixing the previous mistakes. The strongest standard method for tables like ours. Tuned settings: *depth 3, regularisation* $\lambda = 1.0$*, minimum child weight 5*
LightGBM	boosted trees	Same family as XGBoost with a faster algorithm. A speed comparison, often slightly better on wide data. Tuned settings: *leaves 31, minimum child samples 40, regularisation* $\lambda = 3.0$

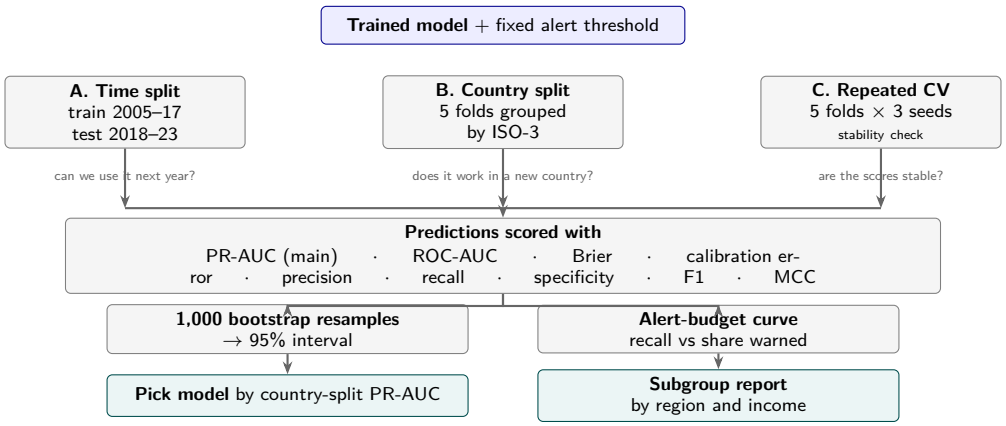


Figure 1: Evaluation pipeline. One trained model is scored three different ways, each answering a different question, then reported with uncertainty intervals.

in the glossary. In short, PR-AUC is the main one because rare yes rows are what we care about, ROC-AUC is reported so others can compare, and Brier plus calibration error check whether the percentages are believable.

Five things stayed inside the training data: model settings, the alert threshold, the calibration step, the missing-value fill, and the choice of which columns to drop. The test years could not touch any of them. Every input comes from the current year or earlier, with only the answer taken from the following year.

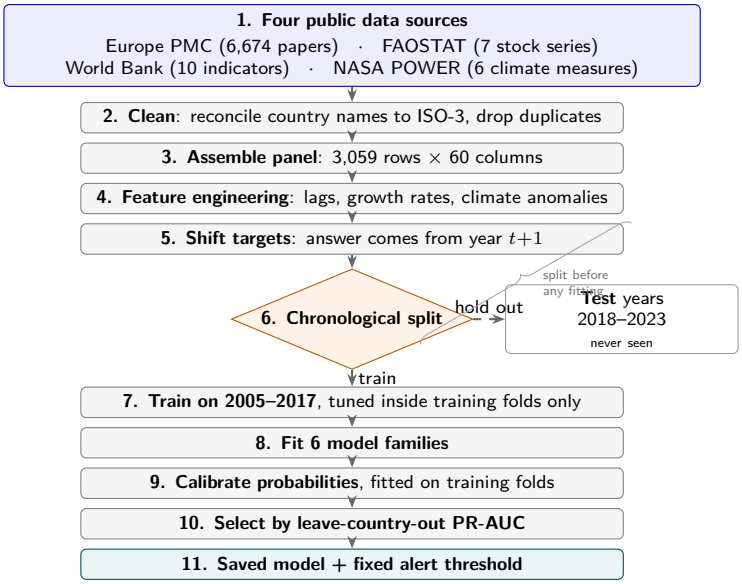


Figure 2: Training pipeline. The chronological split sits before any model sees the data, so the 2018–2023 years never influence training. Tuning and calibration both happen strictly inside the training years.

8 How The Maths Works

Four ideas carry most of the analysis.

Precision and recall. The model flags some rows as risky. *Precision* asks what share of the flagged rows were real, and *recall* asks what share of the real events we flagged. They pull against each other, since flagging more rows raises recall and lowers precision. *F1* combines the two. For an alert system this trade-off is the whole design decision: a missed outbreak costs lives, and crying wolf costs attention, because owners stop reading.

Why PR-AUC instead of accuracy. Sort every row by predicted risk, walking from most to least risky, and at each step compute precision and recall. The area under that curve is PR-AUC, which answers how quickly picking the riskiest rows surfaces real events. A method with no skill scores the prevalence, 0.056 here. The 2,906 easy negative rows cannot inflate it, which is why it suits rare events. ROC-AUC has the opposite weakness: the large negative class makes a mediocre model look strong. We report it so our numbers can be compared with other papers.

Calibration. A ranking can sort countries, but a message quoting a "30% chance" needs that number to be true. We group predictions into buckets and compare each bucket's average prediction with how often the event happened. The averaged gap is the expected calibration error, or 0.025, which is 0.025 here. The correction is fitted on training data only, and it never reorders rows, only the numbers on the axis.

SHAP. SHAP comes from cooperative game theory. If each feature were a player contributing to a prediction, what is each player's fair share of the credit? Fair here means averaging the contribution over every possible order in which features could be added. One property makes it far more useful than simpler measures:

The contributions add up exactly to the gap between the model's average prediction and this particular prediction.

That property is called local accuracy. It lets us say that a prediction is 54% rather than the 5.6% average, and give the exact breakdown of why, with nothing left over.

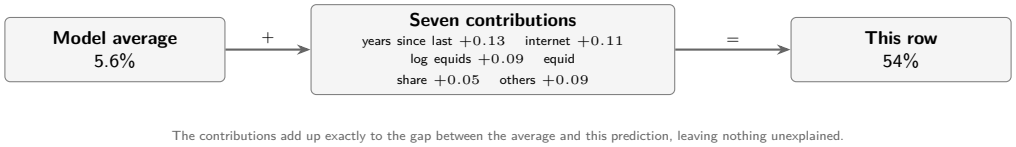


Figure 3: SHAP attributes one prediction to its inputs. Because the contributions must sum exactly to the gap, the explanation is arithmetic rather than a plausible story.

SHAP distributes credit *inside* the model. When two features carry the same information the split between them is arbitrary, and a large value is not evidence of cause.

9 Results

Table 5: Main results. Every column except Precision and Recall is a 0 to 1 score where higher is better. The first four columns come from the time split, where the model is tested on later years. The fifth repeats the main score under the harder country split, where the model is tested on unseen countries. Precision and Recall describe what happens at the alert threshold on the time split.

Model	PR-AUC (time split)	95% interval on PR-AUC	ROC-AUC (time split)	PR-AUC (country split)	Precision (time split)	Recall (time split)
Logistic regression	0.471	[0.36, 0.59]	0.851	0.411	0.55	0.41
Decision tree	0.439	[0.33, 0.56]	0.857	0.241	0.39	0.54
Random forest	0.540	[0.43, 0.65]	0.866	0.318	0.48	0.50
Extra trees	0.551	[0.44, 0.66]	0.868	0.304	0.58	0.47
XGBoost	0.465	[0.35, 0.59]	0.853	0.290	0.64	0.43
LightGBM	0.501	[0.38, 0.61]	0.844	0.315	0.68	0.31

The two PR-AUC columns measure the same thing on different tests. Time split trains on 2005 to 2017 and tests on 2018 to 2023. Country split trains without some countries and tests on those, which is harder and closer to real use. Neither is a precision score.

PR-AUC sits at 0.056 for a model with no skill. On the time split the best model reaches 0.551; on the country split, which is closer to real use, the logistic regression leads at 0.411. The uncertainty column is wide: with 74 positives in the test years a score can move a long way within that range, so neighbouring models should not be separated with confidence.

The ensembles lead on the time split, but on the country split the logistic regression is ahead, and the single tree comes last on both. With only fourteen inputs the relationships are close enough to additive that a regularised linear model transfers best.

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